

Qatar-1b

R-2834

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260801 · created 2026-09-05T23:49:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461253.8928 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.032
Transit depth (Rp/Rs)^2	1.15 +/- 0.69 [%]
Orbital Inclination (inc)	84.36 +/- 2.26
Airmass coefficient 1 (a1)	1.107 +/- 0.019
Airmass coefficient 2 (a2)	-0.073 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.76 %
Best Comparison Star	#1 - [256, 419]
Optimal Method	PSF photometry
Transit Duration (day)	0.063 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.107 ± 0.032 vs 0.1463 (deviation 1.23σ)
Duration fit vs published	0.063 d vs 0.0692 d (deviation 0.39σ)
Mid-time O–C	4.7 min
Depth SNR	3.3
Residual scatter	1.76 %

Light curve

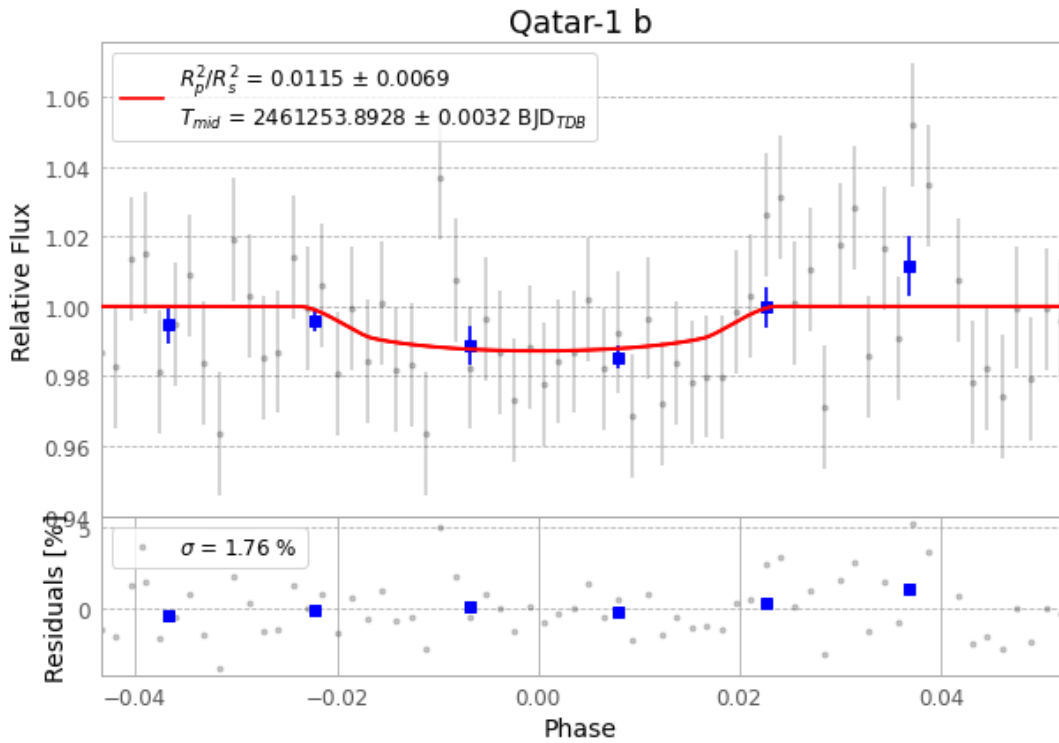


Figure 1 — FinalLightCurve_Qatar-1 b_2026-08-01.png (SHA-256 bc52f615e3c41d58...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 167], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a7d9a28a79391e09...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), Qatar-1260801075415.FITS → Qatar-1260801110915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260801.log (01dd0e71cb12...), FinalLightCurve_Qatar-1 b_2026-08-01.png (bc52f615e3c4...), FinalLightCurve_Qatar-1 b_2026-08-01.csv (c1f5a2aabec7...)

Compute resources

Wall time 146.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:49Z.